Solution Hints

1. x_i = Total production quantity in each week;

   I_i = Inventory items in each week;

   O_i = Quantity of item due to Overtime in each week

   (i= 1…4)

   Minimize $10\,(x_1 + x_2) + 15(x_3 + x_4) + 3(I_1 + I_2 + I_3 + I_4) + 15O_2 + 20O_3$

   Subject to:

$$x_1 - I_1 = 300$$

$$I_1 + x_2 + O_2 - I_2 = 700$$

$$I_2 + x_3 + O_2 - I_3 = 900$$

$$I_3 + x_4 - I_4 = 800$$

$$x_i \leq 700; \qquad i = 1 \dots 4$$

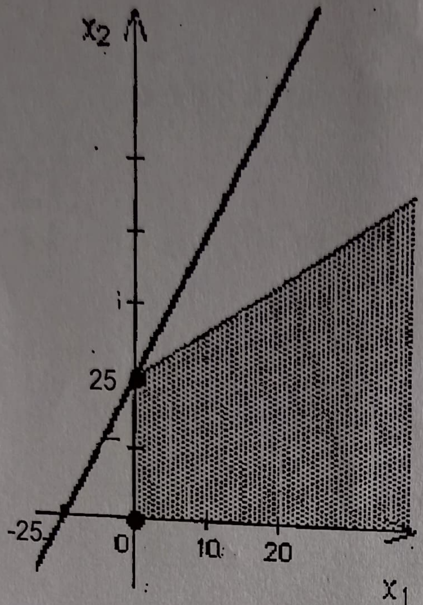
$$x_2 + O_2 \leq 900$$

$$x_3 + O_3 \leq 900$$

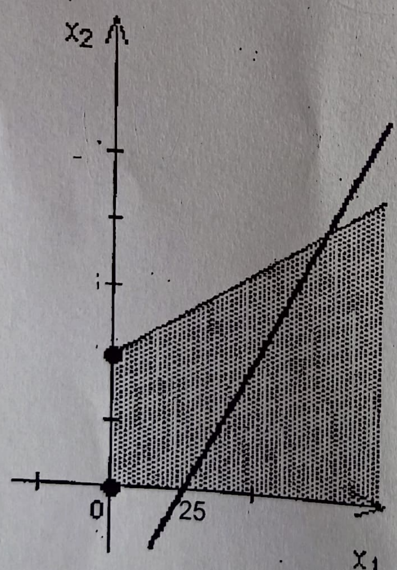
$$x_i, I_i, O_i \geq 0 \qquad i = 1 \dots 4$$

2 a) The feasible region is unbounded.

b) Yes. Optimal Solution: $(x_1, x_2) = (0, 25)$ and $Z = 50$



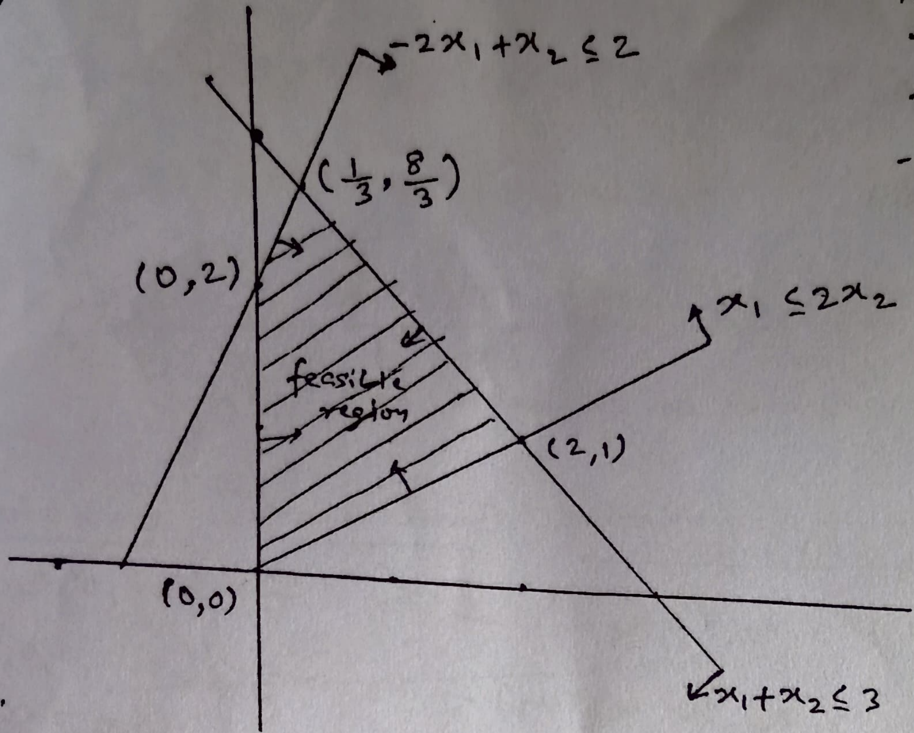
c) No. The objective function value is maximized by sliding the objective function line to the right. This can be done forever, so there is no optimal solution.



d) No, solutions exist that will make Z arbitrarily large. This usually occurs when a constraint is left out of the model.

(3)

(a)



Augmented form

$x_1+x_2+x_3=3$

$-2x_1+x_2+x_4=2$

$x_1-2x_2+x_5=0$

$x_1, x_2, \dots x_5 \ge 0$

(b)

CPF Solⁿ (x_1,x_2)	Corresponding BFS (x_1,x_2,x_3,x_4,x_5)
$(2,1)$	$(2,1,0,5,0)$
$(0,0)$	$(0,0,3,2,0)$

x_3 becomes a basic variable

$\Rightarrow$ Possible Entering variable $= x_3$

Possible Leaving variable $= x_1$ or x_2

BFS	Basic variables	Non-basic variables
$(2,1,0,5,0)$	(x_1, x_2, x_4)	(x_3, x_5)
$(0,0,3,2,0)$	(x_2, x_3, x_4) or (x_1, x_3, x_4)	(x_1, x_5) or (x_2, x_5)

4)

Augmented form

$$\max\ z = 2x_1 + x_2 + 4x_3 + 5x_5 + x_6$$

$$\text{s.t.}\quad 3x_1 + 6x_2 + 3x_3 + 2x_4 + 3x_5 + 4x_6 + x_7 = 60$$

$$x_j \geq 0,\ \forall j = 1,2,\ldots,7$$

maximum no. of basic sol^n $= {}^7C_6 = 7$

S.N.	Non-basic variables (set to zero)	Basic variable	Basic feasible sol^n
1	$(x_2, x_3, x_4, x_5, x_6, x_7)$ —	x_1 —	$(20,0,0,0,0,0,0)$
2	$(x_1, x_3, x_4, x_5, x_6, x_7)$ —	x_2 —	$(0,10,0,0,0,0,0)$
3	$(x_1, x_2, x_4, x_5, x_6, x_7)$ —	x_3 —	$(0,0,20,0,0,0,0)$
4	$(x_1, x_2, x_3, x_5, x_6, x_7)$ —	x_4 —	$(0,0,0,30,0,0,0)$
5	$(x_1, x_2, x_3, x_4, x_6, x_7)$ —	x_5 —	$(0,0,0,0,20,0,0)$
6	$(x_1, x_2, x_3, x_4, x_5, x_7)$ —	x_6 —	$(0,0,0,0,0,15,0)$
7	$(x_1, x_2, x_3, x_4, x_5, x_6)$ —	x_7 —	$(0,0,0,0,0,0,60)$

BFS	1	2	3	4	5*	6	7
z	40	10	80	0	100	15	0

optimal sol^n: $(0,0,0,0,20,0,0)$

$z = 100$

5

since the value of Z can be improved by increasing the value of x_2, the current sol^n is not optimal.

Basis	x_1	x_2	x_3	x_4	x_5	RHS	Ratio
x_3	0	1	1	0	-2	20	20 →
x_4	0	1	0	1	-1	40	40
x_1	1	0	0	0	1	40	–
Z	0	-2	0	0	3	120	
x_2	0	1	1	0	-2	20	
x_4	0	0	-1	1	1	20	20 →
x_1	1	0	0	0	1	40	40
Z	0	0	2	0	-1	160	
x_2	0	1	-1	2	0	60	
x_5	0	0	-1	1	1	20	
x_1	1	0	1	-1	0	20	
Z	0	0	1	1	0	180	

optimal sol^n : $x_1 = 20$, $x_2 = 60$, $x_5 = 20$

$x_3 = x_4 = 0$

$Z = 180$

22204 (Mid term solution)

Q.6 Let $X_1 = x_1^+ - x_1^-$

$\max -Z = -x_1^+ + x_1^- - X_2 - M\bar{x}_4 - M\bar{x}_6$

$-Z + x_1^+ - x_1^- - X_2 \quad + M\bar{x}_4 \quad + M\bar{x}_6 \quad = 0 \quad - ⓪$

$x_1^+ - x_1^- + X_2 - X_3 + \bar{x}_4 \quad = 2 \quad - ①$

$x_1^+ - x_1^- + 2X_2 \quad + X_5 \quad = 8 \quad - ②$

$2x_1^+ - 2x_1^- + 3X_2 \quad + \bar{x}_6 \quad = 10 \quad - ③$

Putting eq⓪ in proper gaussian elimination form

eq⓪ − eq①*M − eq③×M

$-Z + (1-3M)x_1^+ - (1-3M)x_1^- + (1-4M)X_2 \quad + MX_3 = -12M$

BV	x_1^+	x_1^-	X_2	X_3	$\bar{X}_4$	X_5	$\bar{X}_6$	RHS
$-Z$	$1-3M$	$-1+3M$	$1-4M$	M	0	0	0	$-12M$
$\bar{X}_4$	1	-1	1	-1	1	0	0	2
X_5	1	-1	2	0	0	1	0	8
$\bar{X}_6$	2	-2	3	0	0	0	1	10

X_2 is the entering BV

Q.7

ⓐ

BV	X_1	X_2	X_3	X_4	RHS
Z	0	0	0	3	12
X_3	0	1/2	1	$-1/2$	2
X_1	1	1/2	0	1/2	2

ⓑ Current solution is optimal as coeff ⓪ eq ≥ 0

ⓒ Multiple optimal Solution as coeff of NBV X_2 in 0^{th} equ^n is 0. other alternative optima is also degenerate.